Spatial Singularity-Exponent-Domain Multiresolution Imaging-Based SAR Ship Target Detection Method

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***Abstract*— A novel spatial singularity-exponent-domain multiresolution imaging (SSMRI) method is proposed in this article. It is aimed at improving the image processing ability for the single-channel synthetic aperture radar (SAR), thereby acquiring the robust image feature for SAR target detection at the very low SNR. Combining the 2-D singularity power spectrum (SPS) and the 2-D pseudo-Wigner–Ville distribution (PWVD), the SSMRI method is derived. Consequently, the single-channel SAR images are transformed to obtain spatial multiresolution SAR images with respect to the singularity exponent. Furthermore, the separability of the SAR image target in the space of the singularity exponent is analyzed, and the separability theorem in the sense of SSMRI is proved. In particular, under the background of GWN and fractal noise, the separability of SAR image targets based on SSMRI processing is studied. In addition, the stability and robustness of SSMRI-SAR feature extraction are demonstrated. On this basis, a maritime SAR ship target detection method based on SSMRI is put forward in the extremely low SNR condition. The experiment results on the SAR-Ship-Dataset indicate that the proposed method is superior in performance to the traditional CFAR or 2-D-SPS method. Especially when SNR = −30 dB, the detection performance with more than 99.4% can be achieved.**

***Index Terms*— Singularity power spectrum (SPS), spatial singularity exponent domain multiresolution imaging (SSMRI), synthetic aperture radar (SAR), target detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) has all-day and all-weather observation capability and is one of the main means for sea monitoring and target detection. Nowadays, the SAR maritime target detection technology is a key technology for the analysis, processing, and application of massive SAR remote sensing data. However, due to the special imaging mechanism of SAR and the influence of speckle noise, the traditional optical image target processing, and detection methods are not suitable for SAR images. Under the complex marine background, the SAR small target detection is still the key technology to be solved urgently in the field of radar [1]–[3].

To solve this problem, the methods commonly used at present include four categories.

1) One is to improve the resolution of the SAR image. Such high-resolution image processing is used to improve the signal-to-noise ratio (SNR) and the performance of the target detection. However, the means are limited by the technology level of the sensor itself, the observation platform, and the hardware conditions. Thus, it is difficult for SAR images to achieve the resolution level of the visible light and the laser sensors. Nevertheless, there are still a large number of low-resolution SAR image applications.
2) A variety of sensors and observation data are fused to improve the detection performance of SAR small and weak targets. For example, there is the fusion of SAR and optical image [4]–[6], and the fusion of SAR and infrared image [7], [8].
3) The SAR image-processing algorithm, including the feature extraction and detector [9]–[11], can be improved. For example, the target features in the time domain, the frequency domain, and the transformation domain or joint domain are extracted and utilized. The constant false alarm rate (CFAR) detection is commonly used in the SAR target detection based on the threshold optimization with the statistical analysis of the noise background.
4) The artificial intelligence and deep learning methods are adopted to replace the traditional feature extraction and the detector [12], [13], such as densely connected end-to-end neural network (DCENN), Faster-RCNN, YOLOv1, the coupled convolutional neural network, and the spatial shuffle-group enhance attention [14]–[16].

The high performance of the SAR target detection is realized. It is based on a large number of the labeled samples, high-performance computing resources, and iterative optimization of the network.

Among the aforementioned methods, we mainly focus on Category 3) to improve the detection performance of the SAR weak and small targets by the reimaging and the feature extraction. In the traditional feature extraction methods,

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the SAR image processing in the frequency domain, the time–frequency domain, the sparse domain, and so on is utilized to acquire the texture and structure feature. On this basis, the targets' detector is implemented by the maximum likelihood probability or the maximum posterior probability criterion. However, the preceding methods encounter a challenge at the low SNR and complex background. In addition, it is difficult to obtain an ideal detection performance. The early researches indicate that the radar sea clutter behaves as obvious nonstationary, non-Gaussian, and nonlinear characteristics at the low incident angle. The fractal theory [17]–[20] well describes the nonlinear and self-similar characteristics of the complex marine background [17]. Currently, fractal modeling and signal processing methods [21] have brought new opportunities for SAR image target detection. Details may refer to [22]–[24].

At present, the fractal theory-based SAR small target detection mainly includes three approaches.

1) The target detection methods based on the fractal features, for example, the fractal dimension (FD) and multifractal spectrum (MFS) [25]–[29]. However, feature extraction based on fractal parameters has many disadvantages, such as poor feature stability and serious sensitivity to the SNR. In fact, these problems seriously affect the performance of the corresponding detection method.
2) The target detection methods based on the time-singularity MFS distribution (MFSD). MFSD extends the MFSs in the time dimension and provides the time-varying fractal features [30], [31]. The MFSD effectively improves the characterization ability of the dynamic fractal spectrum, but it still has a similar shortcoming to the FD and the MFS. Meanwhile, the MFS and the MFSD are only analyzed from the differentiability and the signal/image structure, which cannot represent the power distribution characteristics of radar signals or images.
3) The target detection methods based on the singularity power spectrum (SPS), such as radar target detection based on the SPS and SAR target detection based on the 2-D-SPS [32], [33]. This method combines the power spectrum and the MFS and effectively improves the detection performance of SAR small targets [34]. However, the SPS method lacks the representation of the spatial power characteristics, and its characterization ability of the localized SPS is insufficient. Therefore, the SPS can only give the overall SPS curve of signal or image but cannot describe the SPS of the local or specific point. Furthermore, the SPS cannot give the power distribution characteristics in the image space with the different singularity exponents. For this reason, it is necessary to combine the advantages of fractal theory and the traditional time–frequency analysis to improve the performance of the target detection.

To solve the abovementioned problems, Li *et al.* [35] put forward the SAR image target detection based on the SPS in the time–frequency domain. It follows from [35] that, in this article, the spatial singularity-exponent-domain multiresolution imaging (SSMRI) method and the SAR small target detection based on SSMRI are proposed, which is derived from the 2-D-SPS and the traditional time–frequency distribution. Major contributions of this article include the following.

1) The definition of SSMRI is proposed. With the theoretical derivation, the multiresolution SAR image analysis is given in detail. The separability theorem of the SAR image based on the SSMRI, with three corollaries in GWN, fractal noise, or multifractal noise, is proved. The SSMRI method provides the spatial multiresolution SAR images with the visual intuitiveness at the specific singularity exponent.
2) The SSMRI method realizes the statistical power spectrum analysis of the local points in SAR image in the singularity exponential dimension, and it is of great value for the feature extraction of the different scattering regions (targets, scenes, and so on), the target edge segmentation, and the target classification.
3) The small dependence of SSMRI on the noise is demonstrated. Furthermore, the SAR target detection method based on SSMRI is proposed at a very low SNR. At SNR $= -30$ dB, the detection probability of more than 99.4% can be achieved with the false alarm rate equal to 10^{-2}.

The rest of this article is organized as follows. Section II introduces the basic idea, the derivation, and the implementation of the SSMRI method. Also, the separability theorem of SAR image target in the sense of SSMRI is proved. In Section III, an SAR target detection method based on the SSMRI method is presented. In Section IV, the experiment results on the SAR-Ship-Dataset verify the performance of the proposed method. Finally, this article concludes with a brief summary in Section V.

II. Spatial Singularity-Exponent-Domain Multiresolution Imaging Method

A. Principle and Definition of the SSMRI Method

Fig. 1 shows the schematic of SSMRI. The SSMRI method includes three parts: the spatial 2-D-PWVD estimation of SAR-SPS estimation and spatial multiresolution imaging in the singularity exponent domain. First, using the 2-D PWVD estimation, the single-channel SAR image is extended from the 2-D image to the 4-D variables, where every pixel of SAR image is mapped to a 2-D frequency-domain subimage (2-D-FDSI). Then, for each "subimage," the 2-D SPS can be, thus, estimated. In this way, the 2-D-FDSI is transformed into the 1-D SPS. Thereby, the spatial spectrum distribution function with three variables (the 2-D spatial variables and the singularity exponential variable) can be acquired. Furthermore, the SSMRI of SAR images, called SSMRI-SAR images, can be achieved.

The SSMRI method connects the power spectrum, the singularity exponent, and the space of the SAR image together so that the SSMRI-SAR images can describe the power distribution of the original SAR image in the joint domain of the singularity exponent dimension and the spatial dimension.

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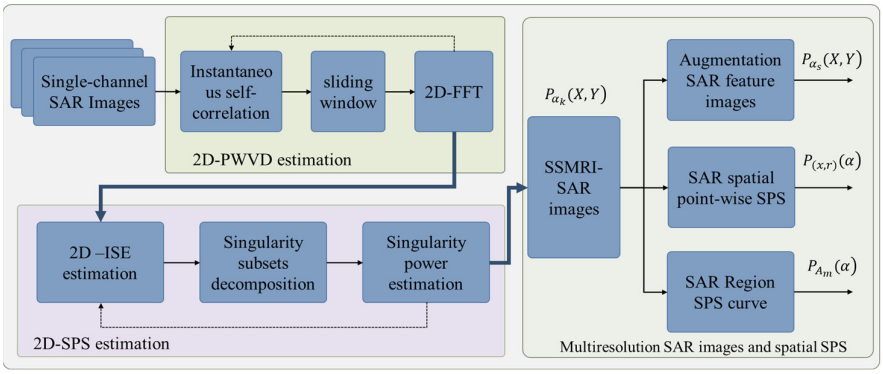


Fig. 1. Schematic of SSMRI.

Among them, the spatial dimensions of SSMRI-SAR images retain the spatial structure of the original image, the SPS ensures the visual intuitiveness of SSMRI-SAR images, and the singularity exponent dimension represents the differentiability and structural characteristic of SSMRI-SAR images.

1) Definition (Spatial Singularity-Exponent-Domain Multiresolution Imaging): For the SAR image $I(x, y)$, I denotes the gray value at the spatial coordinates (x, y). Then, the SSMRI operator $F : I(x, y) \rightarrow P(x, y, \alpha)$ can be defined as

$$P(x, y, \alpha) = F(I) \triangleq \hat{P}\left(\widehat{S}_{(x,y)}\left(\tilde{W}_{(x,y,f_x,f_y)}(I)\right)\right), \quad (1)$$

where $\tilde{W}_{(x,y,f_x,f_y)}(\cdot)$ denotes the 2-D-PWVD operator, $\widehat{S}_{(x,y)}(\cdot)$ denotes the estimation operator of the 2-Dinstantaneous singularity exponent (ISE), and $\hat{P}(\cdot)$ is the estimation and the renormalization of the 2-D-SPS.

These three parts are illustrated in detail in the following.

B. Theoretical Derivation of SSMRI

1) 2-D-PWVD Estimation of SAR Image: The spatial frequency spectrum of the original SAR image can be obtained by the Fourier transform (FT) of the 2-D instantaneous autocorrelation function. Here, we adopt the pseudo-Wigner–Ville distribution (PWVD) to avoid the serious cross term of multicomponents. The discrete form of 2-D-PWVD can be rewritten by

$$\begin{aligned}\mathrm{PW}_f(n_1, n_2, f_1, f_2) &= \sum_{m=-N_2+1}^{N_2-1} \sum_{l=-N_1+1}^{N_1-1} h(m, l) f(n_1 - k, n_2 - l) \\ &\quad \cdot f^*(n_1 + m, n_2 + l) e^{-j2\pi(mf_1/N_1 + lf_2/N_2)}, \end{aligned} \quad (2)$$

where $h(m, l)$ denotes the spatial window function, and here, the rectangle window function is adopted.

From (2), one has the window function of size $(2N_1 - 1) \times (2N_2 - 1)$, and the 2-D FT is performed on each spatial point (n_1, n_2) to obtain the 2-D-FDSI.

2) 2-D-SPS Estimation for the 2-D-FDSI: The 2-D-PWVD of the image realizes the transformation from the SAR image space (x, y) to the 4-D space (x, y, f_x, f_y), that is, each spatial pixel of the original SAR image corresponds to a 2-D-FDSI in the 2-D-PWVD spectrum, which shows the power distribution of the spatial points of the SAR image in two frequency dimensions. Using the 2-D-SPS to the 2-D-FDSI from the 2-D-PWVD spectrum, the SPS of the spatial point can be, thus, obtained. By traversing all the spatial pixels (x, y), the SPS of each spatial point in the SAR image can be acquired. For each SPS curve, clearly, the singularity exponent can be aligned spatially to obtain the feature image of the SAR image, which reflects the information of both the original SAR image and the SPS, thereby providing a richer feature for the SAR image processing.

Now, let $I(x, y)$ denote the original SAR image. Also, let (x_i, y_j) be the pixel, where $x = \{x_1, x_2, \ldots, x_I\}$ and $y = \{y_1, y_2, \ldots, y_J\}$. The discrete 2-D-PWVD of $I(x, y)$ can be represented as

$$\begin{aligned}W(x, y, f_x, f_y) &= \sum_{k=1}^{N_1} \sum_{l=1}^{N_2} h(m, l) I(x - m, y - l) \\ &\quad \cdot I^*(x + m, y + l) e^{-j2\pi\left(\frac{mf_x}{N_1} + \frac{lf_y}{N_2}\right)}, \end{aligned} \quad (3)$$

where $W(x, y, f_x, f_y)$ is the 2-D-PWVD of $I(x, y)$, m and l represents the delay in the x- and y-axes, respectively, and (f_x, f_y) is the 2-D frequency component.

For an SAR image, one takes a pixel $I(x, y)$ as the center and selects an identity matrix of size $N_1 \times N_2$ as the window function. With the aid of (3), the 2-D-PWVD is performed. Then, one slides the window to the next adjacent pixel until all spatial pixels in the image are traversed. Hence, one obtains the 2-D-PWVD of the entire image. Finally, let $W_{x,y}(f_x, f_y)$ be the 2-D-FDSI, i.e., the 2-D-PWVD of the pixel (x, y). Then, we have a matrix with the form

$$W_{x,y}(f_x, f_y) = \begin{bmatrix} W_{x,y}^{f_{x1} f_{y1}} & W_{x,y}^{f_{x1} f_{y2}} & \cdots & W_{x,y}^{f_{x1} f_{yN_2}} \\ W_{x,y}^{f_{x2} f_{y1}} & W_{x,y}^{f_{x2} f_{y2}} & \cdots & W_{x,y}^{f_{x2} f_{yN_2}} \\ \cdots & \cdots & \cdots & \cdots \\ W_{x,y}^{f_{xN1} f_{y1}} & W_{x,y}^{f_{xN1} f_{y2}} & \cdots & W_{x,y}^{f_{xN_1} f_{yN_2}} \end{bmatrix}, \quad (4)$$

where $W_{x,y}^{f_{x_i} f_{y_j}} = \sum_{m=1}^{N_1} \sum_{l=1}^{N_2} I(x - m, y - l) I^*(x + m, y + l) e^{-j2\pi((mf_i/N_1) + (lf_j/N_2))}$.

After the 2-D-PWVD $W(x, y, f_x, f_y)$ or the original SAR image is obtained, the ISE of the 2-D frequency component $W_{x,y}(f_x, f_y)$ can be estimated as the following steps.

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1) For any point (f_x, f_y), select a square area around it, where the size of the square is $\varepsilon_i \times \varepsilon_i$, $\varepsilon_i = 2i - 1$, $i = 1, 2, 3, 4$.
2) For each square area in Step 1), calculate the maximum measure in the area as $\mu_i(f_x, f_y) = \max(W_{x,y}(f_x, f_y))|_{(f_x,f_y)\in\varepsilon_i}$.
3) Define the singularity exponent of each square area as $\alpha_{x,y}(f_x, f_y) = \lim_{\varepsilon_i\to\infty}((\log(\mu_i(f_x, f_y)))/(\log(\varepsilon_i)))$. Taking the limit yields the local singularity exponent. Using $\alpha_{x,y}(f_x, f_y)$, one can construct the singularity set $W^{\alpha}_{x,y}(\eta_x, \eta_y)$ and estimate the SPS as the following two steps.
   a) For any pixel, construct the singularity set Ω based on $\alpha_{x,y}(f_x, f_y)$. Then, determine the maximum value $\alpha_{\max} = \max(\alpha_{x,y}(f_x, f_y))$ and the minimum value $\alpha_{\min} = \min(\alpha_{x,y}(f_x, f_y))$ for $\alpha_{x,y}(f_x, f_y) \in \Omega$, respectively.
   b) Divide $[\alpha_{\min}, \alpha_{\max}]$ with equal interval into K subinterval, that is, $[\alpha_k - \alpha, \alpha_k + \alpha]$ for $k = 1, 2, \ldots, K$, where α_k is the central value of the kth subinterval, and then, the kth singularity subset is composed of all the elements in $W_{x,y}(f_x, f_y)$ satisfying $\alpha_{x,y}(f_x, f_y) \in [\alpha_k - \alpha, \alpha_k + \alpha]$, denoted as $W^{\alpha}_{x,y}(\eta_x, \eta_y)$.
4) The SPS for $W^{\alpha}_{x,y}(\eta_x, \eta_y)$ can be calculated with

$$\begin{cases} P_{x,y}(\alpha_k) = \dfrac{1}{N_\alpha}\sum \left|W^{\alpha}_{x,y}(\eta_x, \eta_y)\right|^2, \\ \alpha_{x,y}((f_x, f_y)) \in [\alpha_k - \Delta\alpha, \alpha_k + \Delta\alpha] \\ \qquad\qquad \text{for } k = 1, 2, \ldots, K, \end{cases} \tag{5}$$

where N_α denotes the number of elements (or the cardinality) of the set $W^{\alpha}_{x,y}(\eta_x, \eta_y)$.

For all spatial pixel points in the original SAR image, using the 2-D-SPS algorithm on the 2-D-FDSI, the SPS $P(x, r, \alpha)$ can, thus, be obtained.

3) Multiresolution Singularity Exponent SAR Image and Regional Statistical SPS Analysis: For $P(x_i, y_j, \alpha_k)$, we can interpret from two perspectives in the following: the first perspective is analyzed in the singularity exponent domain. By cutting the 3-D space in the singularity exponent domain, the SSMRI-SAR images $P_{\alpha_k}(X, Y)$ with different singularity exponents can be obtained, i.e.,

$$P_{\alpha_k}(X, Y) = \begin{bmatrix} P_{\alpha_k}(x_1, y_1) & P_{\alpha_k}(x_1, y_2) & \cdots & P_{\alpha_k}(x_1, y_J) \\ P_{\alpha_k}(x_2, y_1) & P_{\alpha_k}(x_2, y_2) & \cdots & P_{\alpha_k}(x_2, y_J) \\ \vdots & \vdots & \vdots & \vdots \\ P_{\alpha_k}(x_I, y_1) & P_{\alpha_k}(x_I, y_2) & \cdots & P_{\alpha_k}(x_I, y_J) \end{bmatrix}. \tag{6}$$

For $P_{\alpha_k}(X, Y), k = 1, 2, \ldots, K$, select the interval or the set α_s of the singularity exponents. The SSMRI-SAR images $P_{\alpha_s}(X, Y)$ with the singularity exponent resolution can be constructed. That is,

$$P_{\alpha_s}(X, Y) = \sum_{\alpha_k\in\alpha_s} P_{\alpha_k}(X, Y)/\#\{\alpha_s\}. \tag{7}$$

In the application of image detection and target classification, to reduce the influence of noise or scene interference, the selection of the set or interval of singularity exponent can be used to enhance the characteristics of the target area and suppress scene interference.

Since the different scenes or targets have different fractal characteristics, after the SSMRI processing, those characteristics will have different manifestations in different SSMRI-SAR images. In some regions of singularity exponent, the target feature will be enhanced, while, in others, the target feature will be suppressed. Therefore, the appropriate singularity exponent values or intervals α_s should be selected according to the specific requirements of target detection and image classification.

Studies show that the MFS can not only well reflect the local fractal characteristics of SAR image but also reflect the change of the overall characteristics with the ISE. Therefore, the MFS of the SAR mage has important physical significance. For example, $f(\alpha) = 1$ and $f(\alpha) = 2$ represent the texture mutation area in the image, such as the edge and the smooth area of the image, respectively. Also, $0 < f(\alpha) < 1$ and $1 < f(\alpha) < 2$ represent relatively isolated points and texture structure at different levels in the SAR image, respectively. Therefore, in accordance with the mapping relationship $f(\alpha) \to \alpha$, one can select the appropriate ISE value or interval to extract the features of the smooth surface, image edge, image texture, and an outlier in the original image, respectively.

The second perspective is to be analyzed in view of the pixel points of the spatial SAR image. The SPS curves $P_{[x,r]}(\alpha)$ of different scenes can be traced and obtained as follows:

$$P_{(x,y)}(\alpha) = \begin{bmatrix} P_{(x,y)}(\alpha_1) & P_{(x,y)}(\alpha_2) & \cdots & P_{(x,y)}(\alpha_K) \end{bmatrix}. \tag{8}$$

Aiming at the SPS curves of each pixel in the SAR image space, a scene-related SPS curve with a homogeneous region can be constructed based on SSMRI-SAR images with scene labels. First, the typical region segmentation is carried out according to the scene and M kinds of region segmentation images A_m are obtained, i.e.,

$$I(x, y) = \cup_m A_m. \tag{9}$$

The pixels in each region A_m are classified into a specific scene, such as the vegetation A_1, the desert A_2, the cultivated land A_3, the sea surface A_4, the building A_5, and the hull A_6. The spatial pixel subscript set B_m of images is constructed for each scene, i.e.,

$$B_m = \left\{(x, y), \forall I_{x,y} \in A_m\right\}. \tag{10}$$

Thereby, the specific SPS of a given scene region can be obtained as follows:

$$P_{A_m}(\alpha) = \begin{bmatrix} P_{B_m}(\alpha_1), & P_{B_m}(\alpha_2), & \cdots, & P_{B_m}(\alpha_K) \end{bmatrix}, \tag{11}$$

where $P_{B_m}(\alpha_1)$ is the statistical average of the SPS for pixels in the same region at α_1, in which it represents the power measurement of a specific SAR scene or target on a given singularity exponent, namely,

$$P_{B_m}(\alpha_i) = \sum_{(x,y)\in B_m} P(x, y, \alpha_i)/\#\{B_m\}. \tag{12}$$

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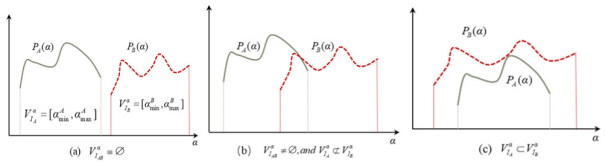


Fig. 2. Relationship between the separability coefficient and the regional SPS of A and B. (a) $V_{I_A}^{\alpha} = \emptyset$. (b) $V_{I_{AB}}^{\alpha} \neq \emptyset$, and $V_{I_A}^{\alpha} \not\subset V_{I_B}^{\alpha}$. (c) $V_{I_A}^{\alpha} \subset V_{I_B}^{\alpha}$.

C. Separability Theorem for SSMRI-SAR Images

Definition 2 (Separability in the Sense of SSMRI): Suppose that there exist the target region A and the background region B in image I. The SSMRI-SAR images of I can be acquired based on the SSMRI processing. Assume that there is some SSMRI-SAR image $P_\alpha(X, Y)$ with the ISE α, satisfying that the power distribution of the background region equals zero, and the power distribution of the target region equals nonzero, namely, $P_\alpha(I_B) \equiv 0$ and $P_\alpha(I_A) \neq 0$. Then, in the sense of SSMRI, the SAR image target has the separability or is separable.

Theorem 1 (Separability Theorem of SAR Image Target Based on SSMRI): Assume that there is only a single target A and the background B in SAR image, namely, $I(x, y) = I_A \cup I_B$, where A and B are evenly homogeneity. The SSMRI-SAR images $P(x, y, \alpha)$ can be obtained by SSMRI processing. Furthermore, the statistical SPS can be computed by means of the pointwise statistical analysis of regions A and B in SSMRI-SAR images, denoted as $P_{I_A}(\alpha)$ and $P_{I_B}(\alpha)$. If it satisfies

$$V_{I_A}^{\alpha} \backslash V_{I_A}^{\alpha} \cap V_{I_B}^{\alpha} \neq \emptyset \tag{13}$$

where $V_{I_A}^{\alpha} = [\alpha_{\min}^{A}, \alpha_{\max}^{A}]$ and $V_{I_B}^{\alpha} = [\alpha_{\min}^{B}, \alpha_{\max}^{B}]$ are the dispersive interval of the singularity exponent (SISE) of $P_{I_A}(\alpha)$ and $P_{I_B}(\alpha)$, respectively, then the target A has complete separability under the background of B, called the SAR target separable for short.

Proof: Let us suppose that $V_{I_{AB}}^{\alpha} = V_{I_A}^{\alpha} \cap V_{I_B}^{\alpha} \neq \emptyset$, and $\tilde{V}_{I_A}^{\alpha} = V_{I_A}^{\alpha} - V_{I_{AB}}^{\alpha} \subset V_{I_A}^{\alpha}$, respectively. It follows from (14) that, if $\tilde{V}_{I_A}^{\alpha}$ is nonempty, one has

$$\tilde{V}_{I_A}^{\alpha} \cap V_{I_B}^{\alpha} = (V_{I_A}^{\alpha} - V_{I_{AB}}^{\alpha}) \cap V_{I_B}^{\alpha} = V_{I_{AB}}^{\alpha} - V_{I_{AB}}^{\alpha} = \emptyset. \tag{14}$$

Thus, one has $\tilde{\alpha} \in V_{I_A}^{\alpha}$ and $\tilde{\alpha} \notin V_{I_B}^{\alpha}$ for any $\tilde{\alpha} \in \tilde{V}_{I_A}^{\alpha}$. This means that $P_{\tilde{\alpha}}(I_B) \equiv 0$ and $P_{\tilde{\alpha}}(I_A) \neq 0$. Therefore, it meets the SAR target separability in the sense of the SSMRI. The proof is therefore complete.

Furthermore, define the separability coefficient of SAR target to be

$$\eta = 1 - V_{I_{AB}}^{\alpha} / V_{I_A}^{\alpha} = 1 - (V_{I_A}^{\alpha} \cap V_{I_B}^{\alpha}) / V_{I_A}^{\alpha}. \tag{15}$$

In Theorem 1, if $V_{I_{AB}}^{\alpha} \equiv \emptyset$, then $\eta = 1$. This implies that both A and B are fully separable (detectable) [see Fig. 2(a)]. In that case, target A and region B are displayed in different SSMRI-SAR images with different ISEs, and the separation (detection) of the SAR target can be achieved from any SSMRI-SAR images $P_\alpha(x, y)$, where $\alpha \in V_{I_A}^{\alpha}$.

If $V_{I_{AB}}^{\alpha} \neq \emptyset$ and $V_{I_A}^{\alpha} \not\subset V_{I_B}^{\alpha}$, then the separability coefficient of the SAR target meets $0 < \eta < 1$ [see Fig. 2(b)], where there exists a similar fractal SISE between target A and region B.

If $V_{I_A}^{\alpha} \subset V_{I_B}^{\alpha}$, as depicted in Fig. 2(c) and the SISE of target A completely covered by region B, then the separability coefficient η is equivalent to zeros. It means that the SAR target is detectable in the sense of SSMRI.

According to Theorem 1 and the separability coefficient, the corollaries under different noise scenarios are derived as follows.

Corollary 1 (Case of Gaussian White Noise): If region B in Theorem 1 can be modeled by Gaussian white noise, theoretically, $P_B(\alpha)$ degenerates to an isolated point at $\alpha = 0$, i.e.,

$$P_B(\alpha) = \begin{cases} c, & \alpha = 0 \\ 0, & \text{elsewhere,} \end{cases} \tag{16}$$

where $V_{I_{AB}}^{\alpha} \equiv \emptyset$, and $c(0)$ is the noise power. In this case, the SAR target A meets the condition of complete separability, i.e., $\eta = 1$.

Corollary 2 (Case of Monofractal Noise): If the region B in Theorem 1 can be modeled by monofractal noise, theoretically, $P_B(\alpha)$ degenerates to an isolated point at $\alpha = H$, namely,

$$P_B(\alpha) = \begin{cases} c_H, & \alpha = H \\ 0, & \text{elsewhere,} \end{cases} \tag{17}$$

wherein H is the Holder exponent of fractal noise and c_H is the fractal noise power. At this point, regardless of whether there is an intersection $V_{I_A}^{\alpha} \cap V_{I_B}^{\alpha}$, the SAR target separation-based SSMRI-SAR images can be achieved by simply avoiding the corresponding ISE, and the separability coefficient is given by

$$\eta = 1 - (V_{I_A}^{\alpha} \cap V_{I_B}^{\alpha}) / V_{I_A}^{\alpha} \approx 1. \tag{18}$$

Corollary 3 (Case of Multifractal Noise): If region B in Theorem 1 can be modeled by the multifractal models, as depicted in Fig. 2(b), it satisfies the separability of the SAR target in the sense of SSMRI with $0 \leq \eta \leq 1$. It is worth noting that, when $\eta = 0$, it does not mean that the SAR target is inseparable in the sense of SSMRI, but the SAR target cannot be directly separated from SSMRI-SAR images. The detection performance is affected by the actual SNR of SSMRI-SAR images corresponding to the singularity index in the cross-region.

Corollaries 1–3 can be obtained directly from Theorem 1 and definitions. Furthermore, extending Theorem 1 to the multitarget and the complex scenario, one has the following Theorem 2.

Theorem 2: Generally, the SAR image $I(x, y)$ can be divided into mutually nonoverlapping N homogeneity connected regional targets A_i for $i = 1, 2, \ldots, N$ and the complex scene B, i.e., $I(x, y) = \cup_{i=1}^{N} I_{A_i} \cup I_B$, where I_{A_i} and I_B are the image values corresponding to the regional coordinate A_i and the complex scene B, respectively. Now, $P(x, y, \alpha)$ denotes the SSMRI of SAR images, and $P_{A_i}(\alpha)$ represents the statistical SPS for the region A_i for $i = 1, 2, \ldots, N$. Now, define $I_{\tilde{A}_i} = \cup_j I_{A_j}, i \neq j, A_j \cap A_i \neq \phi$. If $V_{I_{A_i}}^{\alpha} \backslash V_{I_{A_i}}^{\alpha} \cap (V_{I_{\tilde{A}_i}}^{\alpha} \cup V_{I_B}^{\alpha}) \neq \emptyset$, then the target I_{A_i} is separable

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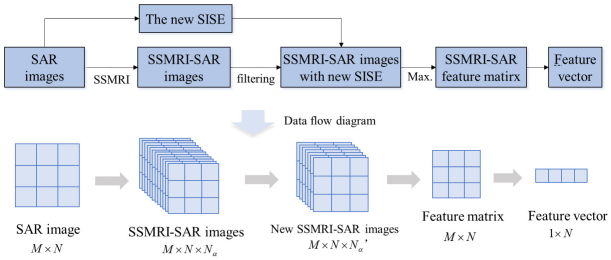


Fig. 3. SAR image feature extraction based on the SSMRI method.

(or detective) in the sense of SSMRI, called I_{A_i}-separability for short. Here, $V^{\alpha}_{I_{A_i}} = [\alpha^{A_i}_{\min}, \alpha^{A_i}_{\max}]$, $V^{\alpha}_{I_{\tilde{A}_i}} = [\alpha^{\tilde{A}_i}_{\min}, \alpha^{\tilde{A}_i}_{\max}]$, and $V^{\alpha}_{I_B} = [\alpha^{B}_{\min}, \alpha^{B}_{\max}]$ represent the SISE of $P_{A_i}(\alpha)$, $P_{\tilde{A}_i}(\alpha)$, and $P_B(\alpha)$, respectively.

The proof of Theorem 2 directly follows from the generalization of Theorem 1.

In that case, the separability coefficient of the SAR target I_{A_i} in the sense of SSMRI is

$$\eta = 1 - \frac{V^{\alpha}_{I_{A_i}} \cap (V^{\alpha}_{I_{\tilde{A}_i}} \cup V^{\alpha}_{I_B})}{V^{\alpha}_{I_{A_i}}}. \tag{19}$$

According to Theorem 1, as long as (13) is satisfied, the SAR image targets are separable in the sense of SSMRI. If $\eta = 1$, it meets the strong separability condition. It is obvious that it is the case for the background of Gaussian noise or monofractal noise. Under the condition of separability, the SAR target can be, thus, detected completely.

III. SAR Target Detection Method Based on SSMRI

In this section, a novel maritime SAR target detection method is proposed based on the SSMRI processing, including the SSMRI-SAR image feature extraction and the support vector machine (SVM) detector.

A. Image Feature Extraction Based on SSMRI-SAR

The SSMRI-SAR images contain abundant information, including the local SPS curves and the space SPS distribution image in the singularity exponent domain. Based on the SPS as given in Section II-B, one can estimate the SISE of SAR image with the target or the pure sea surface and removes their intersection from the SISE of SAR image with the target. To remove the interference of sea surface noise and clutter, the new SISE can be used to filter SSMRI-SAR images to obtain SSMRI-SAR images with the new SISE. Studies show that the maximum value of the column vector in SSMRI-SAR images with the new SISE is a distinguishing feature, and it is sufficient to distinguish the SAR image with and without the targets.

The specific process for the features extraction from the single-channel SAR image slices using SSMRI is depicted in Fig. 3. The steps needed in the process are given as follows.

1) Input SAR image data and calculate the image of size $M \times N$.
2) Estimate the SSMRI of SAR image to obtain the SSMRI-SAR images of size $M \times N \times N_\alpha$, where N_α denotes the number of the discrete SE parameter.
3) Estimate the SISE of SAR image with and without the target, compute the new SISE, and obtain the new SSMRI-SAR images within the new SISE.
4) Find the maximum of the new SSMRI-SAR images in the singularity exponent domain to obtain the SSMRI-SAR feature image of size $M \times N$.
5) Select the maximum value of the column vector of the SSMRI-SAR feature image as the characteristic vector of size $1 \times N$.

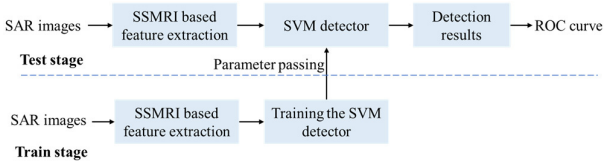


Fig. 4. Process of SAR image target detection based on the SSMRI method.

B. Detector-Based SSMRI and SVM

The process of SAR image target detection based on SSMRI and the SVM is shown in Fig. 4. The real SAR image data are divided into the training set and the test set. Using SSMRI for each SAR image, one can get the corresponding SSMRI-SAR images and the feature vector for the training. The training dataset is used to train and optimize the SVM detector, including the number of the support vectors (SVs) and the penalty factor. Then the optimized SVM detector is utilized in the detection experiment with the SAR image test dataset to observe and validate the detection performance.

IV. Experiment and Result Analysis

This section shows the experimental results to validate the effectiveness of the proposed SAR target detection method based on the SSMRI processing, including the comprehensive detection performance, the antinoise performance, and the calculation efficiency analysis, as well as the experimental summary.

A. Ship SAR Image Dataset [36]

The Ship SAR image slices used are derived from the SAR-Ship-Dataset, which is an open Dataset from the Key Laboratory of Digital Earth and the Chinese Academy of Sciences. The dataset covers the Gaofen-3 SAR sensor and Sentinel-1 sensor. The ship SAR images in the dataset have diverse target types and high complex background, including 43 819 warship SAR image slices, which meets the experimental requirements for the detection of small and small targets under the complex background.

In the SAR-Ship-Dataset, GaoFen-3 sensors have five kinds of imaging modes with the resolution of 3, 5, 10, 8, and 25 m, respectively. The imaging mode of the Sentinel-1 sensor includes the stripe mode (SM) and the interferometry wide (IW) range mode, with the resolution of 1.7 m × 4.3 m to

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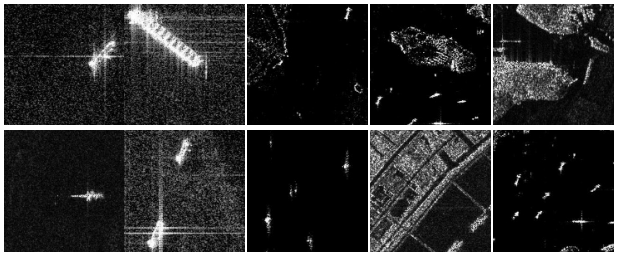


Fig. 5. SAR image samples in the SAR-Ship-Dataset.

TABLE I
SAR IMAGE SAMPLE INFORMATION OF THE SHIP TARGET DATASET

Sensors	Width (km)	Incidence (degree)	Polarization	Scene number
GF-3	30	20-50	single	12
GF-3	50	19-50	dual	20
GF-3	30	20-41	all	20
GF-3	100	19-50	dual	30
GF-3	40	20-38	dual	20
Sentinel-1	80	20-45	dual	98
Sentinel-1	250	29-46	dual	10

TABLE II
DETECTION RESULTS OF DIFFERENT KERNEL FUNCTIONS OF THE SVM

Evaluation	Gaussian function	Linear function	Polynomial function
AUC	1.0000	1.0000	0.5625
GR	0.0089	0.0179	0.4630

3.6 m × 4.9 m, and 20 m × 22 m, respectively. The slice size of the SAR image is 256 × 256, as shown in Table I and Fig. 5.

B. Experimental Setting and Results

1) Data Selection: The 600 SAR image slices, including 300 pure sea surface and 300 ship targets, are selected from the SAR-Ship-Dataset, and the image slice size is 256 × 256. The selective samples are divided into the training set and the test set with the ratio of 8:2.

2) Parameters Setting of SSMRI: In the SSMRI processing, the number of fractal subsets and the sliding window are selected as $N_\alpha = 20$, and $N_1 = N_2 = 13$ after considering the various factors, such as the singularity exponent resolution, the feature extraction ability, and the computation load comprehensively. Thus, the feature matrix of size 256 × 256 is constructed by using the SSMRI processing in which each row corresponds to one observation sample and each column corresponds to the different feature values.

3) Selection of SVM Parameters: The type of the kernel function (the Gaussian kernel function, the linear kernel function, and the polynomial function), the size of the kernel function, and the penalty factor are selected by the experiments and optimization analysis.

It is obvious that Table II reveals the results of tenfold cross-validation on the validation set with different kernel functions. The generalization rate (GR) is the classification error rate in the tenfold cross-validation. Note that the smaller the GR value is, the better the generalization ability of the model. According to the analysis results, the Gaussian kernel function is chosen as the kernel function of the SVM detector. Second, in order to improve the detection performance, Bayesian optimization is used to select the kernel size and penalty factor of the SVM. The kernel size Kernelscale = 61.668 and the penalty factor BoxConstraint = 23.992 are obtained by minimizing the detection error on the verification set, shown in Fig. 6.

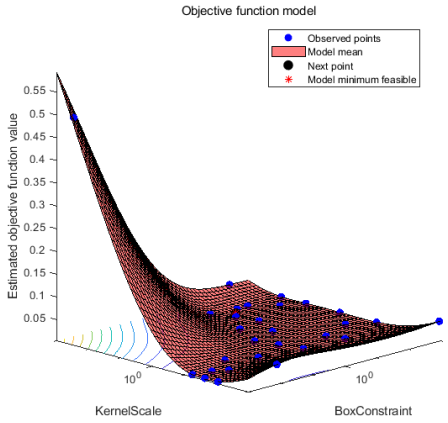


Fig. 6. Hyperparameters and objective function model.

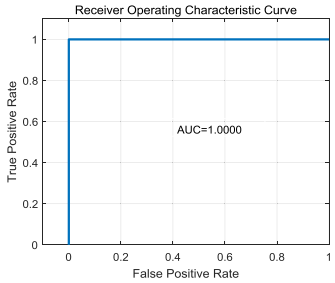


Fig. 7. Detection result after hyperparameter optimization.

Fig. 6 shows the iterative process of the classification error of the objective function with the change of the superparameter. Among them, the *x*-, *y*-, and *z*-axes represent the KernelScale of the kernel function, the penalty factor BoxConstraint, and the minimum estimated classification error of the objective function model. With reference to Fig. 6, the classification error of the objective function gradually decreases and rapidly converges with the continuous iteration of the hyperparameters. It indicates that the SVM model can better fit the maximum classification boundary and has lower GR after the Bayesian optimization.

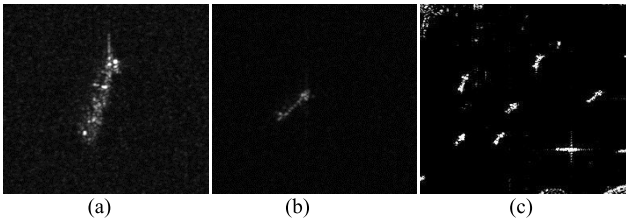
Fig. 7 shows the ROC [35] detection curve of the proposed SSMRI-SAR target detection method on the test set. From the figure, we can find that the AUC value of the proposed method reaches one, which indicates that all pure sea surface SAR images and SAR images of small and small ship targets are correctly detected, and the detection probability is 100%.

Table III shows the detection performance comparison of several detection algorithms with a different false alarm probability (FAP), including the bilateral CFAR (Bi-CFAR) algorithm, the detection algorithm based on the 2-D-SPS method,

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TABLE III
COMPARISON OF THE DETECTION PERFORMANCE

Algorithm / p_d	Detection Probability p_d		
	$p_f = 10^{-1}$	$p_f = 10^{-2}$	$p_f = 10^{-4}$
Bi-CFAR [37]	0.95	0.94	0.75
2D-SPS [34]	0.99	0.97	0.90
SSMRI+SVM	1.00	1.00	1.00



(a) (b) (c)

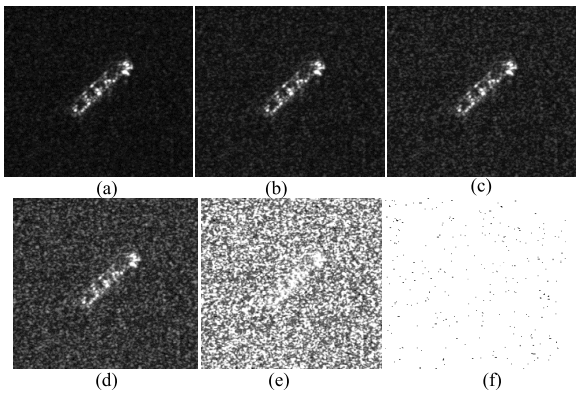
Fig. 8. SAR image slices with different targets. (a) Big target. (b) Small target. (c) Multiple targets.

and the proposed SSMRI-SAR method. Table III also unveils that the Bi-CFAR achieves the detection probability 95% at the FAP, namely, $p_f = 10^{-1}$. However, with the decrease in the FAP, the detection performance decreases significantly. Compared with the Bi-CFAR algorithm, the detection performance of the 2-D-SPS method is greatly improved, especially when the FAP is reduced, i.e., $p_d = 0.90$. The performance of the proposed SSMRI-SAR detection method is obviously better than that of the Bi-CFAR method and the 2-D-SPS method, especially when the FAP decreases.

C. Antinoise Performance Analysis

In order to analyze the robustness of the proposed method at the extremely low value of SNR, the noise (SAR image with pure sea surface) is added to SAR image with ship target to deteriorate the SNR with relative SNR from -30 to 0 dB. Now, assuming that the original image is $I(m,n)$, we add the noise to the original image and get the new image: $I'(m,n) = I(m,n) + \beta \cdot f(m,n)$, where the coefficient is $\beta = ((\|I(m,n)\|^2)/(10^{\mathrm{SNR}/10}\|f(m,n)\|^2)))^{1/2}$, and $f(m,n)$ denotes the SAR image with a pure sea surface. Fig. 8 shows three kinds of SAR target chips, including the large target, the small target, and the multiple target SAR slices. Fig. 9 shows the SAR images with ship target after being added noises with the relative SNR decreasing from 0 to -30 dB. It can be seen from that, when the SNR is less than -5 dB, the SAR ship target image is seriously disturbed by noise. When SNR < -20 dB, the ship target has been completely submerged in sea-surface noise.

Here, 440 SAR image slices used in the experiments, including 300 pure sea surface images, 60 large ship target images, 60 small ship target images, and 20 multitarget images are chosen to be added the noise. Using the mentioned SAR slice, we repeat the SAR target detection experiments based on the proposed SSMRI-SAR method for different relative SNRs. Figs. 10–12 show the ROC curves of the large target, the small targets, and the multitargets. Tables IV–VI show the detection probability with the change of the FAP.



(a) (b) (c) (d) (e) (f)

Fig. 9. SAR image slice for different relative SNRs. (a) 0 dB. (b) -3 dB. (c) -5 dB. (d) -10 dB. (e) -20 dB. (f) -30 dB.

TABLE IV
DETECTION RESULTS OF LARGE TARGETS FOR DIFFERENT RELATIVE SNRS

SNR	Detection Probability p_d	
	10^{-1}	10^{-2}
-30dB	1	0.995
-20dB	1	0.977
-10dB	1	0.981
-5dB	1	0.970
-3dB	1	0.976
0dB	1	0.968

TABLE V
DETECTION RESULTS OF SMALL TARGETS FOR DIFFERENT RELATIVE SNRS

SNR	Detection Probability p_d	
	10^{-1}	10^{-2}
-30dB	0.999	0.997
-20dB	1	0.986
-10dB	1	0.962
-5dB	0.988	0.972
-3dB	1	0.977
0dB	1	0.985

TABLE VI
DETECTION RESULTS OF MULTIPLE TARGETS FOR DIFFERENT RELATIVE SNRS

SNR	Detection Probability p_d	
	10^{-1}	10^{-2}
-30dB	1	0.985
-20dB	1	0.973
-10dB	1	0.965
-5dB	1	1
-3dB	1	0.979
0dB	1	0.982

One can observe from Tables IV–VI that we can find that the SAR target detection method based on the SSMRI method has achieved the ideal detection performance for all three types of targets. When the false alarm rate is 10^{-2}, the average detection probabilities are 97.8%, 98%, and 98%, respectively. In addition, the detection performance is not affected with the relative SNR decreasing. When SNR $= -30$ dB, the detection

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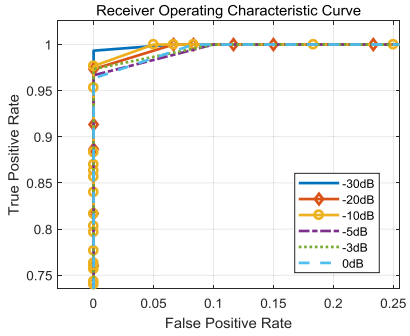


Fig. 10. ROC curves of large targets for different relative SNRs.

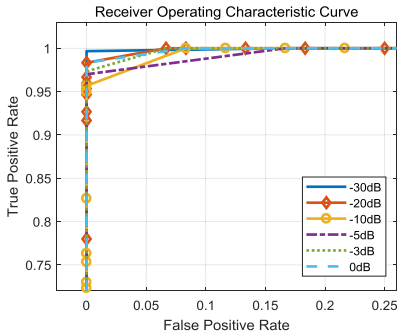


Fig. 11. ROC curves of small targets for different relative SNRs.

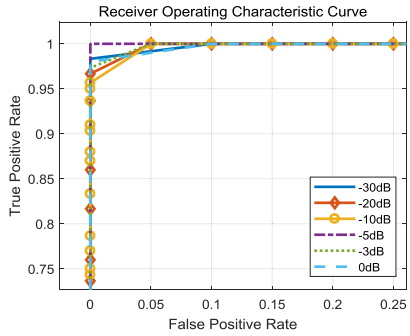


Fig. 12. ROC curves of multiple targets for different relative SNRs.

probabilities are 99.5%, 99.7%, and 98.5%, respectively, and those results are even better than the results at SNR > -20 dB. We conclude that it may be caused by the limited sample size and the randomness of noise.

In order to observe the benefits introduced by the SSMRI processing with the extremely low SNR in more detail, to illustrate this, an example of SAR ship image is given. First, the noise with the relative SNR $= -30$ dB is added to this ship image. Then, we carry out the SSMRI processing to obtain SSMRI-SAR images. Fig. 13 shows the results of SSMRI-SAR images. As can be seen from Fig. 13(b), after adding noise, the target in the original SAR image has been completely submerged. Fig. 13(c)–(h) unveils that the SSMRI-SAR images obtained under different singularity exponents have great differences. When $\alpha = 0$, the SSMRI-SAR images mainly show the random distribution of noise power. When $\alpha = 1.5314$, 3.0628, 4.5942, 6.1255, and 7.6569, from Fig. 13(d)–(h), the SSMRI-SAR image can clearly display the ship target region with high SPS values. The foregoing analysis

TABLE VII

AUC VALUES OF DETECTION RESULTS FOR DIFFERENT RELATIVE SNRS AND TARGET TYPES BY OUR METHOD

AUC	Large target	Small target	Multiple target
-30dB	0.9998	0.9997	0.9992
-20dB	0.9991	0.9994	0.9992
-10dB	0.9994	0.9982	0.9989
-5dB	0.9983	0.9975	1
-3dB	0.9987	0.9991	0.9993
0dB	0.9985	0.9993	0.9990

TABLE VIII

AUC VALUES OF THE THREE METHODS FOR DIFFERENT RELATIVE SNRS [34]

AUC	SSMRI	2D-SPS[35]	CFAR
-30dB	0.9996	0.768	0.312
-20dB	0.9992	0.884	0.422
-10dB	0.9988	0.998	0.482
-5dB	0.9986	0.997	0.650
-3dB	0.9990	/	0.812
0dB	0.9989	0.998	0.960

also proves why the SSMRI method has strong robustness even at very low SNR.

The AUC values of the ROC testing curve for the three kinds of targets are presented in Table VII. From this table, the AUC value of the ROC curve of the proposed method reaches more than 0.99. It indicates that the proposed method has a good antinoise performance, and the AUC values are almost unaffected by the relative SNR. As a comparative analysis, the AUC values of the various methods with the SNR, including 2-D-SPS, CFAR, and SSMRI, are listed in Table VIII. In this table, the AUC value of the proposed method reaches more than 0.99, which indicates that the proposed method has a good antinoise performance, and the AUC values are almost unaffected by the relative SNR. Compared with the 2-D-SPS and CFAR methods, the proposed SSMRI method has better and more robust detection performance at the extremely low SNR.

D. Comparison Analysis With the Deep Learning Method

Furthermore, the proposed SSMRI+SVM detection method is compared with data-driven deep network models, including the Inception V3 model, Faster-RCNN-DR [14], and DCENN [38]. It is worth noting that the SSMRI method proposed in this article is not inconsistent with the DL model, and the deep network model combined with the SSMRI method can also be used. The network structure of the Inception V3 can refer to [39]. The three-channel in the Inception V3 corresponds exactly to three SSMRI images from a single SAR image slice. The learning rate of the deep network is 0.0003, and the batchsize is 32. The relevant parameters of SSMRI are $N_1 = N_2 = 13$ and $K = 6$. Three subimages of SSMRI with $k = 4 \sim 6$ are selected as the input of network training.

Fig. 14 shows the comprehensive detection results based on the proposed methods, where the original SAR images labeled with real ship targets are depicted in Fig. 14(a).

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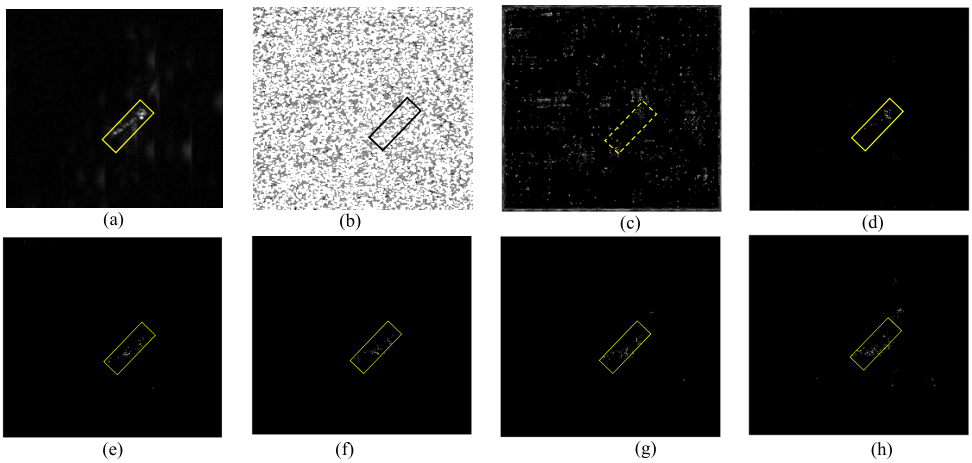


Fig. 13. SSMRI results: (a) original SAR image, (b) noised SAR image at SNR $= -30$, and (c)–(h) SSMRI images with different ISEs. (a) Original SAR image. (b) SNR $= -30$ dB. (c) $\alpha = 0$. (d) $\alpha = 1.5314$. (e) $\alpha = 3.0628$. (f) $\alpha = 4.5942$. (g) $\alpha = 6.1255$. (h) $\alpha = 7.6569$.

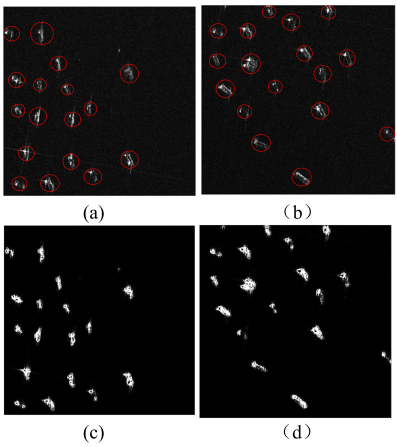


Fig. 14. Original SAR images labeled with ship targets and the detection results with SSMRI+SVM. (a) and (c) Image 1. (b) and (d) Image 2.

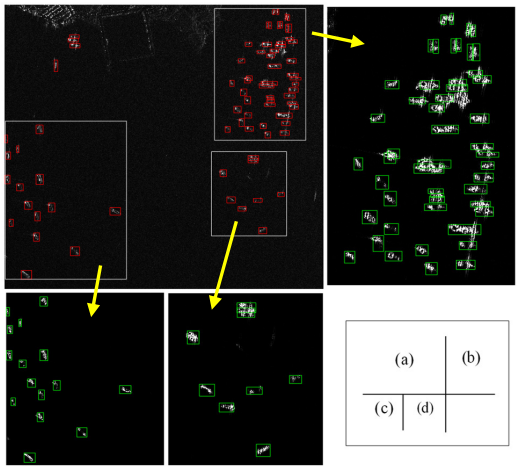


Fig. 15. Detection results with SSMRI+SVM. (a) Original SAR image with bounding-box targets. (b)–(d) Detection results of the three local regions in (a).

The corresponding 34 targets in Fig. 14(c) and (d) are correctly detected with large SNR gain, respectively.

Additionally, an SAR target detection experiment on a larger sea region of size 3000×3000 is carried out, and the results are demonstrated in Fig. 15. The original SAR target image with bounding box is shown in Fig. 15(a), and the detection results of the three target areas in Fig. 15(a) are depicted and magnified in Fig. 15(b)–(d), respectively. We can find that all targets are correctly detected, including the dense and small targets.

As a result, the detection accuracy based on the five methods, including Inception V3, SSMRI+Inception V3, Faster-RCNN-DR, DCENN, and the proposed SSMRI+SVM, has been given in Table IX for comparison analysis, where the results based on Faster-RCNN-DR and DCENN are from the ideas of Sun *et al.* [14] and Jiao *et al.* [38].

We can directly verify from Table IX that: 1) without using SSMRI, the DCENN method performs best with detection accuracy 96.7%, the Inception V3 is 94.9%, and Faster_RCNN DR is worst with 92.9%; 2) the detection accuracy of the Inception V3 is 100%, which is improved by 5.1% after combined with the SSMRI method; and 3) both SSMRI+SVM and SSMRI+Inception V3 achieve 100% detection accuracy on a relatively small dataset. It is worth mentioning that, although the same dataset is used, the detection experiment based on the SSMRI +SVM method and Inception V3 is carried on a small sample, while the detection result in [14] is based on a large sample.

The above experiment further illustrates the advantages of the SSMRI processing in the SAR target detection field, and a significant improvement can be obtained, not only for the traditional SVM method for the data-driven and deep learning model.

TABLE IX

TEST RESULTS BASED ON THE DEEP LEARNING MODELS COMBINED WITH THE SSMRI METHOD

Accuracy	Training data	Testing data
Faster-RCNN-DR [14]	/	92.9%
DCENN [38]	/	96.7%
Inception V3	97.7%	94.9%
SSMRI+ Inception V3	100%	100%
Proposed method	100%	100%

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TABLE X
HARDWARE CONFIGURATION OF SIMULATION EXPERIMENT

CPU	Intel(R) Core(TM) i9-10900K CPU @ 3.70GHz 3.70GHz
RAM	64 GB
GPU	NVDIA GeForce RTX 2080 Ti

TABLE XI
PROCESSING TIME OF SSMRI FOR THE IMAGE SAMPLE OF SIZE 256×256

Running time per image	Length of Sliding Window (Points)				
	9	11	13	15	17
SSMRI-SAR	10.0s	14.9s	21.0s	27.5s	35.6s

E. Computational Efficiency

The hardware configuration of the experimental platform used in the experiment is shown in Table X. Table XI shows the running time required to process a 256×256 image. It is based on SSMRI with different sizes of the sliding window. It can be seen from Table XI that the running time of SSMRI is significantly positively correlated with the sliding window length. Considering the calculation time and the performance, the appropriate length of the sliding window can be selected.

F. Experimental Summary

Based on the abovementioned experiments, the summary is obtained as follows.

1) The proposed SSMRI method and the detection method based on SSMRI-SAR image have a good detection performance for both the maritime SAR weak small target or multiple targets. Compared with the Bi-CFAR method and the 2-D-SPS method, the detection probability is improved by 6% and 3% at $p_f = 10^{-2}$ and 25% and 10% at $p_f = 10^{-4}$, respectively.

2) Compared with the 2-D-SPS method, the experimental results show that the proposed detection method has better antinoise performance and stability. At SNR $= -30$ dB, the AUC value is still beyond 0.999. Meanwhile, for a variety of target types, the detection probability of the proposed algorithm is still about 99.4% with the decreasing of both the FAP and the SNR.

3) The SSMRI can be applied to data-driven SAR target detection methods, and combined with SSMRI, the detection accuracy of Inception V3 is further improved by 5.1%.

Although the proposed method has a significant advantage in detection performance, the efficiency of the algorithm needs to be improved, and it is currently not applicable to the target detection in the large SAR scenario. In the meantime, the influence of parameter setting on the feature extraction and the detection performance of different detection scenes is different to some extent, so it is difficult to obtain a universal method for the parameter setting.

V. CONCLUSION

The SAR image target detection method based on the SSMRI method is proposed in this article. With the aid of SSMRI, the single-channel SAR image can be transformed in the singularity exponent domain to obtain spatial multiresolution SSMRI-SAR images. The separability theorem in the sense of SSMRI is proved. Furthermore, we analyze the separability coefficient with respect to the different noise backgrounds. Research shows that SSMRI processing can extract the multiresolution and subtle features of SAR images at the extremely low SNR. Experiments on the SAR-Ship-Dataset indicate that the proposed method is superior to the traditional CFAR detection method and the 2-D-SPS method. It has a more robust detection performance under the extremely low SNR. The conclusions of this article can be used as a reference for image target detection for the complex background. However, the method proposed in this article has the disadvantage of increasing the computational complexity. For this reason, the efficiency of the proposed algorithm needs to be further improved.

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